STUDENT PERFORMANCE ANALYZER

A Python-Based Data Analytics Dashboard

Project type: Beginner-level Python data analytics project

Technology: Python, Streamlit, pandas, matplotlib, CSV

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1. Abstract

The Student Performance Analyzer is a small data analytics application that helps users review marks across four subjects. It reads a CSV file, calculates totals and average marks, groups results into performance bands, visualizes subject-wise averages and grade distribution, and lists students below a chosen average threshold. The application is intended as a learning and discussion aid, not as an automated decision-making system.

2. Problem Statement

Student marks are often stored in spreadsheets or CSV files. Manually calculating totals, averages, and class-level trends can take time and may lead to mistakes. A simple dashboard can automate routine calculations and present results in a clear visual form.

3. Objectives

• Read student marks from a CSV file. • Validate that required columns are present. • Calculate total and average marks. • Assign understandable performance bands. • Display subject averages and grade distribution charts. • Filter students and export analyzed results.

4. Tools and Technologies

Python provides the application logic. pandas handles tabular data and calculations. Streamlit creates the interactive browser-based interface. matplotlib creates charts. CSV is used as the input and output data format.

5. Methodology

The application loads the sample CSV by default or accepts a user-uploaded CSV. It checks for Student_ID, Student_Name, Maths, Science, English, and Computer columns. Subject marks are converted to numeric values, invalid rows are excluded, and total and average marks are calculated. Each average is assigned a performance band: A for 90 and above, B for 80–89.99, C for 70–79.99, D for 60–69.99, and Needs Support below 60. Finally, metrics, charts, a support list, and a downloadable results file are displayed.

6. System Requirements

Hardware: a laptop or desktop with internet access for installing packages. Software: Python 3.10 or newer, a modern web browser, and the packages listed in requirements.txt. After installation, the app runs locally on the user's computer.

7. How to Run

Extract the project ZIP. Open a terminal in the project folder. Run `python -m pip install -r requirements.txt`, then run `python -m streamlit run app.py`. Open the local URL shown in the terminal if a browser does not open automatically.

8. Expected Output

The dashboard shows the number of students, class average, highest average, and count below the selected threshold. It also shows a table of student marks and calculated values, a subject-average bar chart, a performance-band chart, a support list, and a CSV download option.

9. Limitations

The sample dataset is fictional and small. The tool does not predict future performance, explain the causes of marks, or replace a teacher's judgement. The grade bands are simple configurable rules and may not match an institution's official grading policy. Uploaded files must use the required column names.

10. Future Enhancements

Possible improvements include adding more subjects, importing Excel files, allowing configurable grading rules, adding attendance data, improving chart styling, and adding secure user authentication if the project is deployed for multiple users.

11. Conclusion

This project demonstrates practical beginner-level skills in Python, CSV handling, data validation, aggregation, visualization, and building a small interactive dashboard. It turns raw marks into a more readable summary that can support further discussion and learning.